



A Three-Stage Beamforming with Harmonic Guidance for Multi-Channel Speech Enhancement

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Abstract

With the rapid advancement of multi-channel speech enhancement (MCSE) research, nonlinear spatial filtering methods integrating spatial and spectral processing are prevalent. However, many existing approaches overlook the explicit extraction of speech spectral structure information, leading to insufficient learning of joint spatial-spectral information, limiting performance under low signal-to-noise ratio (SNR) conditions. To address this, we propose a three-stage multi-channel speech enhancement framework. The first stage has an acoustic structure extraction module capturing speech spectral patterns from noisy inputs, enabling spatial-spectral cue exploration and interaction. In the next two stages, the process combines full-band noise reduction with speech structure refinement by decoupling enhancement into coarse enhancement and spectral refinement. Experiments on LibriSpeech-based datasets demonstrate that the proposed method significantly outperforms the reference method.

Index Terms: Multi-Channel Speech Enhancement, Joint Information, Spatial Filtering

1. Introduction

Multi-channel speech enhancement aims to denoise speech signals more efficiently by effectively using spatial cues across channels, enabling better separation of speech from noise [1]. This spatial information, which is not present in single-channel methods, significantly improves the clarity and effectiveness of speech enhancement across various acoustic environments [2, 3].

Neural beamforming, which leverages deep learning to improve traditional beamforming weight prediction, has become popular for its efficiency and generalizability [4, 5]. Typically, these methods follow a two-stage process: predicting time-frequency masks to estimate statistical parameters, followed by deriving beamforming weights from these statistics [6]. While these approaches leverage the spatial selectivity of beamforming, separating spatial processing from temporal-spectral information overlooks their inherent interdependencies. End-to-end frame-wise neural beamforming, which processes spatial and temporal-spectral information simultaneously, has demonstrated superior performance [7–10]. To understand the reason for its effectiveness, in [11], frame-by-frame beamforming is performed using learned embeddings that encapsulate both spatial and spectral discriminative information. This study suggests that a comprehensive spatial-spectral representation is crucial for the network to effectively distinguish between speech and noise patterns. In [12], the authors control the informa-

tion sources available to the network, enabling it to learn combined spatial and temporal-spectral information, showing that the key to improved performance is exploiting the interdependencies between these domains, rather than the non-linear operations of the filtering process. Building on this, [13] further improves performance by introducing additional LSTM layers to extract full-band and sub-band spectral information, enhancing the modeling of spectral structure patterns, though this increases model complexity. However, unlike the above methods for modeling spectral patterns, recent single-channel speech enhancement methods use a priori knowledge to model the speech acoustic structure [14–16].

Given the critical role of spatial-spectral interaction in MCSE performance, researchers have focused on effectively extracting spatial information across channels [17–19]. However, in low-SNR environments, both the inherent acoustic structure and frequency information of speech can be heavily distorted by noise, limiting the network’s ability to explore spectral information and hindering its interaction with spatial information. Inspired by the success of modeling speech acoustic structures in single-channel speech enhancement [15, 16], we propose a three-stage beamforming method with harmonic guidance. Specifically, in the first Feature Extraction stage, we design a Structural-Spectral Embedding (SSE) module to extract blurred harmonic structures from noisy spectrograms as an auxiliary feature. In the subsequent Coarse Enhancement stage, we use harmonic structures as cues to achieve preliminary noise suppression while preserving the acoustic structure. Finally, by utilizing a residual iterative strategy, the joint information is learned in a step-by-step and in-depth manner during each iteration, leading to the continuous optimization of spectral details. Experimental results demonstrate that our model significantly outperforms the baseline.

2. Signal Model

Consider the time-domain noisy and reverberant speech signal captured by the p -th microphone as $x^{(p)}(t)$, where $p = 0, \dots, P - 1$. In the short-time Fourier transform (STFT) domain, the physical model of speech is given by,

$$\mathbf{X}_{t,f} = \mathbf{c}_f S_{t,f} + \mathbf{V}_{t,f} + \mathbf{N}_{t,f} = \mathbf{S}_{t,f} + \mathbf{V}_{t,f} + \mathbf{N}_{t,f}, \quad (1)$$

where $\mathbf{X}_{t,f}, \mathbf{S}_{t,f}, \mathbf{V}_{t,f}, \mathbf{N}_{t,f} \in \mathbb{C}^{P \times 1}$, denote the observed signal, anechoic clean speech, reverberant speech, and noise, respectively, at the time index $t \in \{1, \dots, T\}$ and frequency index $f \in \{1, \dots, F\}$. The first microphone is chosen as the reference channel by convention. The sound sources are assumed to remain static within each speech segment. The goal is to simultaneously denoise and dereverberate the signal to restore anechoic speech.

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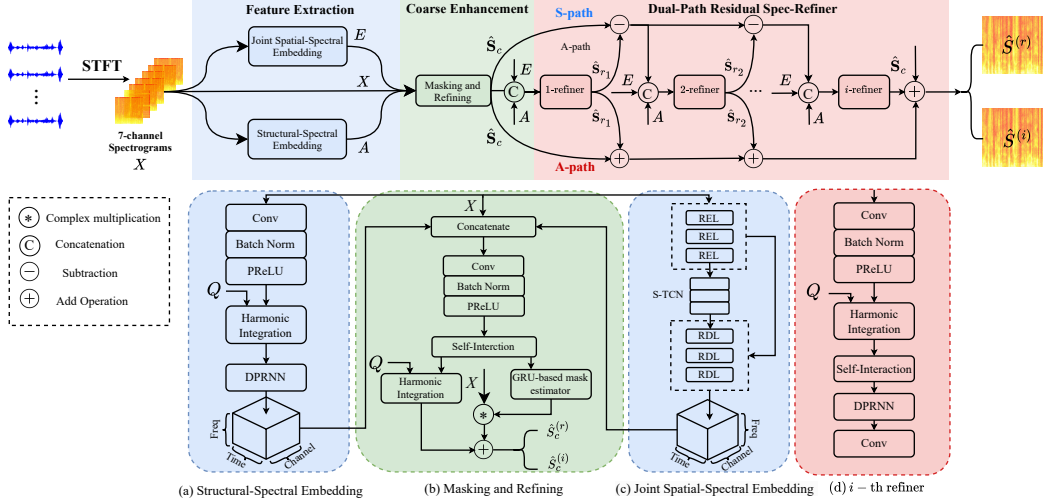


Figure 1: The proposed pipeline consists of three main components: Feature Extraction, Coarse Enhancement, and the Dual-Path Residual Spec-Refiner modules. The Feature Extraction module actively captures spatial and spectral information, generating two feature embeddings. The Masking and Refining module performs coarse enhancement, while the Dual-Path residual Spec-Refiner focuses on fine-grained spectral restoration. Detailed descriptions of each block are provided below the overall framework.

3. Proposed Method

3.1. Overview

The overall diagram of the proposed framework is shown in Fig. 1. The framework consists of Feature Extraction, Coarse Enhancement, and Dual-Path Residual Spec-Refiner (DPRSR). The Joint Spatial-Spectral Embedding (JSSE) in the Feature Extraction module is based on the Embedding network from [11] and, together with the SSE, produces two feature representations that capture joint spatial-spectral information and specific acoustic structures. The following two-stage enhancement network first uses these embeddings, along with the original noisy speech, to perform coarse enhancement through a specially designed multi-input and multi-output Masking and Refining module. Finally, the Dual-Path Residual Spec-Refiner module, composed of multiple cascaded refiners, takes the multi-channel coarse enhancement results along with residuals from the previous refiners. Each refiner progressively refines spectral details that were missed in earlier enhancement steps.

3.2. Structural-Spectral Embedding

Most MCSE models rely on the mixed speech log power spectrum or arrange data differently in Recurrent Neural Networks (RNNs) to capture spectral patterns [13, 20]. However, learning these patterns directly from noisy speech via non-linear mapping may be insufficient, as the acoustic structures of speech (e.g., formants and harmonic cues) are heavily masked by noise in low-SNR environments. This degradation severely limits the network’s capacity to discern and recover critical speech-specific patterns, even with non-linear transformations. Therefore, inadequately learned spectral information has difficulty in interacting effectively with spatial information. Spatial features such as inter-channel phase difference (IPD) and directional feature [21], with structural traits in the time-frequency (T-F) domain, help separate speech from noise. Similarly, we introduce the SSE module to explicitly model spectral patterns by extracting intrinsic acoustic features, particularly the comb-like harmonic structure.

The SSE module consists of a Conv block, Harmonic Integration [15], and DPRNN [22]. The reference channel’s noisy

complex spectrum $\mathbf{X}^{\text{ref}} \in \mathbb{R}^{T \times 2 \cdot F}$ is fed into the Conv block, which consists of a causal 2D convolution (Conv) followed by Batch Normalization and PReLU. The Harmonic Integration (HI) module computes the harmonic distribution via a comb-pitch conversion matrix $\mathbf{Q} \in \mathbb{R}^{L \times F}$ [15] with L pitch candidates:

$$\mathbf{H} = \text{softmax} \left(\text{Conv} \left(\mathbf{X}^{\text{ref}} \right)^2 \cdot \mathbf{Q}^T \right) \cdot \mathbf{Q}, \quad (2)$$

where $\mathbf{H} \in \mathbb{R}^{T \times C \times F}$ indicates the theoretical harmonic distribution at each frequency band. Next, $\mathbf{P}$ highlights the low-energy harmonic structure, which is computed as:

$$\mathbf{P} = \text{Conv} \left(\text{Conv}(\mathbf{X}^{\text{ref}}) \odot \text{Conv}(\mathbf{H}) \right), \quad (3)$$

where $\odot$ is element-wise multiplication. While $\mathbf{P}$ integrates sub-band and full-band spectral information, it lacks temporal modeling, which is addressed by using DPRNN to capture long-term dependencies,

$$\mathbf{A} = \text{DPRNN}(\mathbf{P}), \quad (4)$$

3.3. Masking and Refining

Speech signals exhibit inherent sparsity in the frequency domain [23], often manifesting as periodic harmonic structures. As mentioned in 3.2, these harmonics can serve as additional cues to interact with spatial information. However at the same time, they can also act as specific enhancement targets that align with human auditory perception. Therefore, we designed a Masking and Refining module, which performs full-band noise reduction through masking operations while simultaneously refining the acoustic structure. This allows harmonic information and spatial information to jointly reconstruct the spectral pattern of speech, ensuring the result aligns with human auditory perception through information interaction between these two types of information.

This module takes the embedding $\mathbf{A}$ and the output embedding $\mathbf{E}$ of the JSSE, concatenates them with the noisy speech along the channel dimension, and then reduces the channel dimension using a convolutional block, batch normalization, and

PReLU. The output is fed into a Self-Interaction module, which contains two stacked multi-head Self-attention: one along the channel dimension and the other along the frequency dimension. Self-attention is employed to model dependencies across different dimensions. When working with 3D embeddings that contain spatial-spectral information, Self-attention enables interactions between elements in different dimensions (e.g., frequency and channel), enhancing the model's ability to capture interdependency between spatial and spectral information. This process is expressed as:

$$\mathbf{I} = \text{SI}(\text{BN}(\text{Conv}(\text{Concat}(\mathbf{A}, \mathbf{E}, \mathbf{X})))), \quad (5)$$

where $\text{SI}(\cdot)$, $\text{BN}(\cdot)$, $\text{Concat}()$ represent self-Interaction, batch normalization, and concatenation, respectively. The resulting joint information $\mathbf{I}$ is then passed through a GRU-based mask estimator to predict the complex ratio mask (CRM). The estimator is composed of a GRU module that receives narrow-band data arrangements, with GRU shared across frequency subbands, and fully connected layers (FC). This allows the GRU to capture fine-grained spatial and temporal information while focusing on global spectral statistics. The process is described by:

$$\mathbf{M} = \text{FC}(\text{GRU}(\mathbf{I})), \quad (6)$$

Additionally, $\mathbf{I}$ passes through a Harmonic Integration module that leverages the joint spatial-spectral information in $\mathbf{I}$ to distinguish and enhance the original acoustic structure. This module captures fine-grained spectral information. The overall enhancement process is given by:

$$\hat{\mathbf{S}}_c = \mathbf{M} * \mathbf{X} + \text{HI}(\mathbf{I}), \quad (7)$$

where $\text{HI}(\cdot)$ represents Harmonic Integration, $*$ denotes complex multiplication, $\hat{\mathbf{S}}_c$ is the result of coarse enhancement. After the Masking and Refining process, we believe the model not only restores specific acoustic structures but also effectively removes noise from non-harmonic frequencies, even in the steering direction.

3.4. Dual-Path Residual Spec-Refiner

Decomposing the speech enhancement task into coarse enhancement and subsequent iterative restoration has proven to be effective in both single-channel and multi-channel tasks [24, 25]. However, multi-channel tasks often reduce the coarse enhancement result to a single channel, which sacrifices the integrity of both spatial and spectral processing. To this end, we propose a Dual-Path Residual Spec-Refiner that refines spectral details using multi-channel results and joint spatial-spectral information.

As shown in Fig. 1(b), multiple refiners are connected in series to form a sequential structure. For example, with i refiners, the module receives joint information $\mathbf{I}$ and the residual spectrum results from the previous $i - 1$ refiners, given by $\hat{\mathbf{S}}_c - \sum_{j=0}^{i-1} \hat{\mathbf{S}}_{f_j}$, where $\hat{\mathbf{S}}_{f_j}$ represents the output of the j -th refiner. Unlike single-channel systems, which rely mainly on spectral harmonic features, our module uses joint spatial-spectral information to correct unnatural speech structures. The spatial features within this joint information help guide the model to identify time-frequency bins dominated by speech, including harmonic structures. This process is expressed as:

$$\mathbf{R}_i = \text{HI}\{\text{BN}(\text{Conv}(\text{Concat}(\mathbf{A}, \hat{\mathbf{S}}_c - \sum_{j=0}^{i-1} \hat{\mathbf{S}}_{f_j})), \mathbf{Q}\}, \quad (8)$$

$$\hat{\mathbf{S}}_{f_i} = \text{Conv}(\text{DPRNN}(\text{SI}(\mathbf{R}_i))), \quad (9)$$

In addition, we introduce the A-path and S-path in a residual iterative manner to facilitate collaborative interaction between different modules and progressively further explore spatial and spectral information in the feature embeddings. The S-path, functioning as a residual connection, involves subtracting the output of the j -th refiner, $\hat{\mathbf{S}}_{f_j}$, from the coarse enhancement result $\hat{\mathbf{S}}_c^{(p)}$. This subtraction operation enables each refiner to focus on exploring the spectral and spatial information that was not fully captured during the earlier coarse enhancement and refiner stages. Conversely, the A-path performs an additive operation, gradually restoring and enhancing the spectral details of the coarse enhancement result step-by-step.

4. Experiment Result

4.1. Dataset

We generate multi-channel noisy-clean speech pairs using the LibriSpeech corpus [26] (train-clean-100, dev-clean, and test-clean subsets). Noise data is sourced from the DNS-Challenge [27], with approximately 20,000 samples selected for training. Multi-channel room impulse responses (RIRs) are simulated with a 7-microphone uniform linear array, spaced 5cm apart. Room dimensions range from 5m-5m-3m to 10m-10m-4m (length, width, height), with reverberation times (T60) between 0.1s and 1.0s. To generalize the model, the distance from the target and noise sources to the microphones ranges from 0.5m to 5.0m in 0.5m increments. The direction of arrival (DOA) difference between target and noise is at least 5° . Training data has a SNR randomly chosen from [-6dB, 6dB]. We generate 40,000 training pairs and 4,000 validation pairs, and the duration of each segment is 4 seconds. The test dataset includes about 50 types of environmental noises from MUSAN [28], with four challenging SNR conditions (-10dB, -5dB, -2dB, 0dB), each containing 200 test pairs, specifically chosen to emphasize our focus on enhancement tasks in low-SNR environments.

4.2. Training Setup

All utterances are sampled at 16 kHz. A 20ms Hanning window is used with a 50% overlap between frames, and 320-point FFT is applied. The parameters in the JSSE module are consistent with those in [11], except that the number of channels in the encoder and decoder is set to 32. The channel numbers for the SSE and each refiner are 6, 12 and 32, 14, respectively. The kernel sizes for the convolution in the Conv block and HI are (5,2) and (1,3), respectively, with all convolutions having a stride of (1,1). The HI and Self-Interaction modules employ 4, 8, and 1 heads, respectively. The GRU module is a 2-layer unidirectional GRU with 64 hidden units. JSSE is retrained from scratch rather than fine-tuned. Anechoic speech is used as the reference signal to measure the SI-SDR [29], PESQ [30], and STOI [31] of the enhanced speech.

We introduce a combination of the time-domain scale-invariant signal-to-noise ratio (SI-SNR) and PMSQE [32] based on psychoacoustic perception, as the overall loss function, where the two losses are weighted equally (in a 1:1 ratio).

4.3. Comparison with Baseline

A total of six baselines are selected, including traditional adaptive beamformers with Oracle settings (i.e., Oracle MVDR)

Table 1: *Quantitative comparisons with advanced baselines. The values are specified with PESQ/STOI/SI-SDR formats. Bold face indicates the best performance.*

Method	param.(M)	Signal-to-Noise Ratio				Average
		-10db	-5db	-2db	0db	
Noisy	-	1.41/0.35/-10.59	1.51/0.40/-5.65	1.60/0.43/-2.11	1.60/0.51/-0.42	1.53/0.42/-4.69
Oracle MVDR	-	2.12/0.69/7.55	2.32/0.73/8.10	2.47/0.77/9.00	2.51/0.83/9.99	2.36/0.75/8.66
FaSNet-TAC [7]	2.76	2.37/0.85/5.32	2.67/0.89/7.53	2.81/0.90/8.40	2.90/0.91/8.91	2.69/0.89/7.54
NSF [20]	12.96	2.66/0.87/5.57	2.85/0.89/7.10	2.93/0.90/7.51	2.97/0.91/8.29	2.85/0.89/7.11
FT-JNF [12]	3.35	2.70/0.88/6.11	2.82/0.90/7.19	2.88/0.91/7.73	2.92/0.90/8.81	2.83/0.90/7.46
EaBNet [11]	2.83	3.13/0.93/9.30	3.30/0.94/10.48	3.37/0.95/10.98	3.40/0.95/11.26	3.30/0.94/10.50
TaEr [10]	6.03	2.79/0.89/7.86	2.99/0.90/9.53	3.05/0.90/10.01	3.10/0.91/10.57	2.98/0.90/9.49
Ours	3.21	3.57/0.95/12.30	3.69/0.97/13.49	3.73/0.97/13.93	3.76/0.97/14.21	3.69/0.97/13.48

Table 2: *Ablation study on our Librispeech-based dataset.*

Method	N	Para. (M)	PESQ	SI-SDR (dB)	STOI (%)
Ours w/ N refiners	0	2.39	3.41	10.55	94.31
	1	2.66	3.46	11.19	94.97
	2	2.93	3.49	11.91	95.34
	3	3.21	3.57	12.30	95.98
	4	3.48	3.50	12.11	95.41
EaBNet	-	2.83	3.13	9.30	93.01
+SSE	-	2.95	3.22	10.02	93.86
EaBNet[Large]	-	3.38	3.12	9.27	92.84
Ours	3	3.21	3.57	12.30	95.98
- SSE	3	3.13	3.21	11.57	95.01
- JSSE	3	1.36	3.44	10.95	95.89
JSSE $\rightarrow$ cosIPD	3	1.37	3.47	11.12	94.98

[33], and five outstanding DNN-based approaches. We evaluate these six baseline models on our dataset. As shown in Table 1, under different SNR conditions, our model outperforms the baseline model in all evaluation indicators. From the table, It can be observed that even under low-SNR conditions, the proposed model still maintains its advantage in objective indicators. This is because we introduced acoustic information into the model design and fully integrated it with spatial information. It can also be observed that, similar to our multi-stage model, TaEr, which uses the amplitude-phase decoupling strategy, shows moderate performance in various indicators. This confirms the superiority of our multi-stage model in low-SNR scenarios. Compared to the State-of-the-art (SOTA) model EaBNet, our model achieves 12% and 28% improvements in PESQ and SI-SDR with only 13% more parameters. This demonstrates that explicitly extracted spectral information enhances the model’s ability to learn joint spatial-spectral features in low-SNR conditions. Our system notably improves PESQ due to the incorporation of spectral acoustic structure, particularly harmonics, which aligns better with human auditory perception.

4.4. Ablation Study

Table 2 shows that increasing the number of refiners (N) improves all metrics. With $N = 3$, SI-SDR increases by 1.75 dB compared to no refiners, indicating effective repair of missed spectral details. However, with $N = 4$, performance decreases, suggesting that additional refiners become redundant. Therefore, $N = 3$ is used for baseline comparisons. To verify whether simply increasing the number of parameters improves performance in low-SNR acoustic environments, we increased

the number of channels in EaBNet’s convolution and TCN modules, resulting in a parameter count similar to our model with three refiners (EaBNet[Large]). This, however, degraded performance, highlighting that more parameters do not always lead to better results. In contrast, the advantage of our model in low-SNR acoustic conditions stems from explicitly modeling the acoustic information of the speech itself and integrating multi-channel spatial information. This promotes information interaction and fusion during end-to-end training and leverages harmonics to guide the model to fully learn the joint spatial-spectral information. Integrating the SSE module into EaBNet (+SSE), which uses the extracted spectral information as additional guidance, improves PESQ and SI-SDR by 0.09 and 0.72 dB, respectively. This shows that combining prior knowledge with implicit learning through the nonlinear mapping capabilities of neural networks enhances performance with fewer parameters. Removing the SSE or JSSE modules ($-SSE$, $-JSSE$) reduces the metrics, emphasizing the importance of spatial and spectral information. Without the SSE, PESQ shows a significant drop, confirming the critical role of harmonic structures in speech intelligibility. This also validates our hypothesis that the SSE module can effectively enrich spectral information in the joint spatial-spectral representation learned by the enhancement system, especially under low-SNR conditions. PESQ, an objective speech quality metric, evaluates spectral and temporal features based on human auditory perception. When a neural network learns sufficient spectral information, it can accurately reconstruct speech signals, making enhanced speech closer to the clean signal in frequency and harmonics. This improves naturalness and intelligibility, leading to higher PESQ scores. Replacing JSSE with cosIPD ($JSSE \rightarrow cosIPD$) slightly reduces performance but outperforms EaBNet. This demonstrates the network’s ability to integrate spatial and spectral information, though relying solely on phase information causes some loss.

5. Conclusion

In this paper, we propose a three-stage multi-channel speech enhancement framework addressing the challenges of spatial-spectral information learning in low-SNR environments. The core innovation lies in the acoustic structure extraction module, which captures spectral patterns from noisy signals, effectively integrating spatial-spectral features. Experimental results show that modeling the acoustic structure and interacting it with spatial information can address the information learning bottleneck in multi-channel speech enhancement. This improvement is particularly significant in low-SNR scenarios, where existing methods often struggle to maintain performance.

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